14/10/2026

Code E Phase 5

Aakash
Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18, Udyog Vihar, Gurugram, Haryana - 122015, *Ph.*+91-1244168300

MM : 720

Fortnightly Test for NEET-2027 (XII-Passed_P5)_FT-01E

Time : 180 Min.

Topics Covered:
Physics: Units & Measurements, Motion in a Straight Line, Motion in a Plane
Chemistry: Some Basic Concepts of Chemistry, Structure of Atom
Botany: Cell: The Unit of Life, Cell Cycle & Cell Division, The Living World
Zoology: Structural Organisation in Animals– Animal Tissues, Biomolecules

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

1. Which of the following is a dimensional constant?
(1) Magnification (2) Relative density
(3) Gravitational Constant (4) Strain

2. In a Vernier calipers, one main scale division is equal to N cm and n division of the Vernier scale coincides with $(n-1)$ division of the main scale. The least count (in cm) of the calipers is
(1) $\frac{N}{n+1}$ (2) $\frac{N}{n-1}$
(3) $\frac{N}{n}$ (4) $\frac{Nn}{n-1}$

3. If two physical quantities A and B have different dimensions, then which mathematical operation given below is physically possible?
(1) $\sqrt{AB}$ (2) $A + B$
(3) $A - B$ (4) $B(A + B)$

4. If unit of time is 10 s, unit of mass is 5 kg and unit of length is 20 m. Then the new unit of energy will be
(1) 20 J (2) $\frac{1}{20}$ J
(3) 16 J (4) 40 J

5. The dimensional expression $[ML^2T^{-3}]$ represents
(1) Torque (2) Momentum
(3) Power (4) Pressure

6. Two resistances $R_1 = (100 \pm 2)\ \Omega$ and $R_2 = (200 \pm 4)\ \Omega$ are connected in series. The percentage error in equivalent resistance is
(1) 2% (2) 3%
(3) 4% (4) 5%

7. The value of (24.36 + 0.0623 + 256.2) with proper significant digit is
(1) 280.7 (2) 280.6
(3) 280.622 (4) 281

8. If area (A), velocity (v) and density (ρ) are taken as base units, then dimensional formula of force can be represented as
(1) ρAv (2) $Av^2\rho$
(3) $Av\rho^2$ (4) $A^2v\rho$

9. If $x = \frac{A^2B^3}{\sqrt{C}}$ and the percentage error in the measurement of A, B & C are 1%, 2% & 2% respectively, then maximum percentage error in the value of x would be
(1) 9% (2) 6%
(3) 7% (4) 14%

10. A micrometer has 150 equal divisions on the circular scale, and on one full rotation of circular scale main scale advances by 0.02 cm on the main scale. The least count of the micrometer (in cm) is
(1) 4×10^{-5} (2) 6.21×10^{-3}
(3) 1.33×10^{-4} (4) 1.21×10^{-2}

11. Given below are two statements:
Statement-I: A choice of change in different units does not change the number of significant figures in a measurement.
Statement-II: when two physical quantities are multiplied, there should be as many decimal places retained in the final result as are there in the number with the least decimal places.
In light of above statements, choose the most appropriate answer from the options given below
(1) Both statements I and II are correct
(2) Both statements I and II are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

12. The dimensions of $\frac{a}{b}$ in the relation $P = \frac{a - t^2}{bx^2}$, where P is the pressure, x is the distance and t is the time, is
(1) $[M^{-1}LT^{-2}]$ (2) $[ML^0T^2]$
(3) $[MLT^{-2}]$ (4) $[ML^2T^{-2}]$

13. The value of $(10004)^{1/4}$ according to binomial approximation is
(1) 10.001 (2) 10.04
(3) 10.004 (4) 10.01

14. Which among the following relation cannot be deduced by dimensional method?
(1) $y = A\cos(\omega t - kx)$ (2) $S = ut + \frac{1}{2}at^2$
(3) $K = \frac{1}{2}mv^2$ (4) All of these

15. Match list-I with list-II.

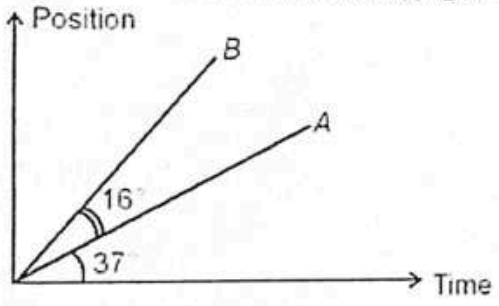
	List-I		List-II
a.	Velocity	(i)	$[M^1L^1T^{-2}]$
b.	Acceleration	(ii)	$[M^0L^1T^{-2}]$
c.	Force	(iii)	$[M^1L^{-1}T^{-2}]$
d.	Pressure	(iv)	$[M^0L^1T^{-1}]$

Choose the correct answer from the options given below:
(1) a(iv), b(ii), c(i), d(iii) (2) a(iv), b(ii), c(iii), d(i)
(3) a(iii), b(ii), c(i), d(iv) (4) a(ii), b(i), c(iii), d(iv)

16. The distance travelled by a particle, starting from rest and moving with a uniform acceleration 4 m/s^2, in 4th second is
(1) 7 m (2) 14 m
(3) 21 m (4) 12 m

17. A car covers half of total distance with speed 30 km/h and remaining half with speed 60 km/h. The average speed for complete journey is
(1) 40 km/h (2) 45 km/h
(3) 50 km/h (4) 60 km/h

18. The position-time graph of two particles A and B are shown in figure. Then ratio of their velocities (v_A /v_B) is



(1) $\frac{3}{4}$ (2) $\frac{4}{3}$
(3) $\frac{9}{16}$ (4) $\frac{16}{9}$

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19. Two cars A and B are moving in straight line opposite to each other with speed 20 m/s and 25 m/s respectively. Magnitude of velocity of car A w.r.t. car B will be
(1) 45 m/s (2) 15 m/s
(3) 5 m/s (4) 20 m/s

20. A ball is thrown vertically upwards with velocity of 20 m/s from top of multistorey building 30 m high. How high will the ball rise above ground (g = 10 ms^{-2})?
(1) 40 m (2) 45 m
(3) 50 m (4) 60 m

21. The graph between position and time, for a particle undergoing uniform motion is
(1) Parabola (2) Straight Line
(3) Hyperbola (4) Ellipse

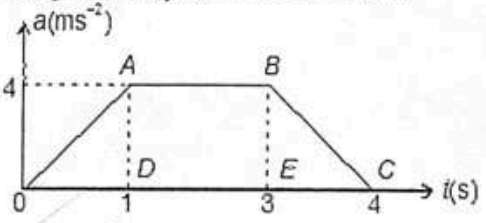
22. The velocity of a particle is given by $v = \sqrt{169 - 8x}$, where x is position in metre and v in m/s. The acceleration of the particle will be
(1) 7 m/s^2 (2) –4 m/s^2
(3) –8 m/s^2 (4) –2 m/s^2

23. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
A : Without mentioning the observer, there is no meaning of rest or motion.
R : Motion of a body is defined only with respect to an observer.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

24. A particle is moving along x-axis so that its velocity is given as $v = (x^2 + x + 4)$ m/s, where x is in m. The acceleration of the particle at x = 1.5 m is
(1) 21 m/s^2 (2) 26 m/s^2
(3) 28 m/s^2 (4) 31 m/s^2

25. The position of a particle moving along x-axis varies with time as $x = t - \sin t$, where x is in metre and t in second. Which of the following value of velocity of the particle is not possible?
(1) 1.6 m/s (2) 0.86 m/s
(3) 1.0 m/s (4) 2.4 m/s

26. For the acceleration-time (a-t) graph shown in figure, the change in velocity from t = 1 s to t = 4 s is

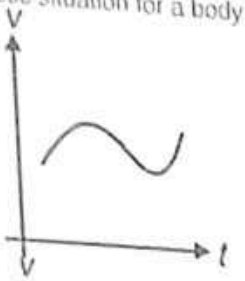


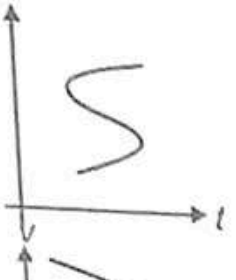
(1) 10 ms^{-1} (2) 12 ms^{-1}
(3) 8 ms^{-1} (4) 16 ms^{-1}

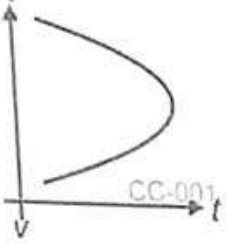
27. A man standing at the top of a tower of 45 m height drops a coin. The velocity with which it hits the ground is (Take g = 10 m/s^2)
(1) 10 m/s (2) 20 m/s
(3) 30 m/s (4) 40 m/s

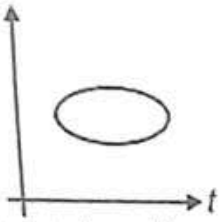
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8. Which of the following velocity(v)-time(t) graph shows a realistic situation for a body in motion?

(1) 

(2) 

(3) 

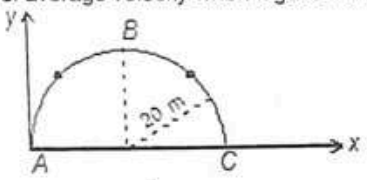
(4) 

29. A body is dropped from a height h above ground. The ratio of distances fallen in first two seconds, next two seconds and next three seconds is
(1) 1 : 3 : 5 (2) 1 : 5 : 7
(3) 1 : 2 : 3 (4) 4 : 12 : 33

30. Two particles move in concentric circular paths of radius r_1 and r_2 such that their angular velocity is same. If the velocity of the first particle is v_1 and for second particle v_2 then the ratio of their time period is
(1) $v_1 : v_2$ (2) 1 : 1
(3) $\frac{r_1}{r_2}$ (4) $\frac{r_2}{r_1}$

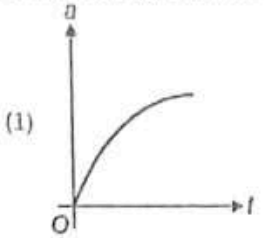
31. A ball is thrown at an angle θ and another ball is thrown at an angle (90°–θ) with horizontal direction from the same point each with speed 30 m/s. The ratio of maximum height attained by them will be.
(1) $\tan^2\theta$ (2) $\sin^2\theta$
(3) $\cos^2\theta$ (4) $\sec^2\theta$

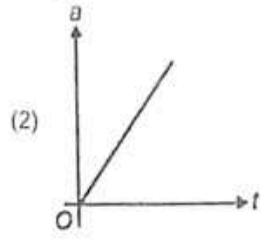
32. A particle travels along a semicircular curve from A to B in 1 s. If it takes 3 s to go from A to C, determine the magnitude of average velocity when it goes from B to C.

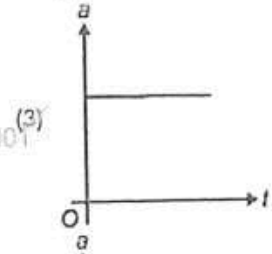


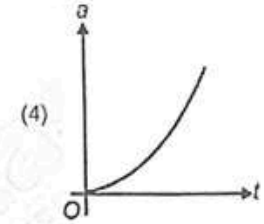
(1) $10\sqrt{2}\ ms^{-1}$ (2) $20\sqrt{2}\ ms^{-1}$
(3) $\frac{10}{\sqrt{2}}\ ms^{-1}$ (4) $\frac{5}{\sqrt{2}}\ ms^{-1}$

33. The velocity v of a particle moving along x-axis varies with its position (x) as $v = \alpha\sqrt{x}$; where α is a constant. Which of the following graph represents the variation of its acceleration (a) with time (t)?

(1) 

(2) 

(3) 

(4) 

34. The trajectory of an object thrown at an certain angle with horizontal and moving under gravity is
(1) Circular (2) Parabolic
(3) Elliptical (4) Hyperbolic

35. A particle moving with uniform speed in a circular path maintains:
(1) Constant velocity
(2) Constant acceleration
(3) Constant velocity but varying acceleration
(4) Varying velocity and varying acceleration

36. River is flowing with velocity $\left(\hat{i}+\hat{j}\right)$ m/s. A boat is moving with velocity $\left(\hat{i}-\hat{j}\right)$ m/s w.r.t. the river. The velocity of boat w.r.t. ground is
(1) $-2\hat{i}$ m/s (2) $2\hat{i}$ m/s
(3) $2\hat{j}$ m/s (4) $2\sqrt{2}\hat{i}$ m/s

37. If angle between two vectors $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{6}$, then angle between $\vec{a}$ and $-2\vec{b}$ will be
(1) $\frac{\pi}{3}$ (2) $\frac{\pi}{6}$
(3) $\frac{2\pi}{3}$ (4) $\frac{5\pi}{6}$

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38. Two bullets are fired horizontally with different velocity from the same height. Which will reach the ground first?
(1) Slower one
(2) Faster one
(3) Both will reach simultaneously
(4) It can't predicted

39. Two projectiles are thrown horizontally from a high cliff in opposite direction with same speed 20 m/s simultaneously At what time velocities of both particles become perpendicular?
(1) $2\sqrt{2}$ s (2) 2 s
(3) 4 s (4) $4\sqrt{2}$ s

40. A cannon can give maximum speed of 70 m/s to bombs. The maximum distance up to which cannon projects a bomb is (g = 10 m/s^2)
(1) 490 m (2) 245 m
(3) 100 m (4) 300 m

41. In the following question, a statement of assertion (A) is followed by a statement of reason (R)
A : A food packet is droped from an aeroplane flying horizontally. The path of food packet is straight line for the pilot.
R : Horizontal velocity of plane and food packet is same
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

42. A man wants to cross a river 600 m wide flowing at 8 m/s in shortest time. Man has a speed of 6 m/s in still water. What is the path length covered by man in the river?
(1) Infinite (2) 1 km
(3) 100 m (4) 1000 cm

43. An object A is moving with constant velocity $\vec{v}_A = \left(5\hat{i} - 2\hat{j}\right)$ m/s and object B is moving with constant velocity $\vec{v}_B = \left(3\hat{i} + 5\hat{j}\right)$ m/s. The velocity of B w.r.t. A is
(1) $\left(2\hat{i} - 7\hat{j}\right)$ m/s (2) $\left(-2\hat{i} + 7\hat{j}\right)$ m/s
(3) $-\left(2\hat{i} + 7\hat{j}\right)$ m/s (4) $\left(2\hat{i} + 7\hat{j}\right)$ m/s

44. A particle starts moving with constant acceleration with initial velocity $\left(\hat{i} + 5\hat{j}\right)$ m/s. After 4 second its velocity becomes $\left(3\hat{i} - 2\hat{j}\right)$ m/s. The magnitude of its displacement in 4 second is
(1) 5 m (2) 10 m
(3) 15 m (4) 20 m

45. When a man walks on a horizontal road with velocity 1 km/h, the rain appears to him coming vertical at a speed of 2 km/h. The actual speed of the rain w.r.t ground is
(1) $\sqrt{3}$ km/h (2) $\sqrt{5}$ km/h
(3) 1 km/h (4) 3 km/h

CHEMISTRY

46. The SI unit of thermodynamic temperature and luminous intensity respectively are
(1) Kelvin and ampere (2) Fahrenheit and candela
(3) Kelvin and candela (4) °C and ampere

47. Match prefixes given in List-I with multiple given in List-II

List-I	List-II
a. mega	(i) 10^{18}
b. giga	(ii) 10^{6}
c. exa	(iii) 10^{12}
d. tera	(iv) 10^{9}

The correct match is
(1) a(iii), b(ii), c(i), d(iv) (2) a(ii), b(iv), c(iii), d(i)
(3) a(iv), b(ii), c(i), d(iii) (4) a(ii), b(iv), c(i), d(iii)

48. Which of the given pairs has the compounds with same empirical formula?
(1) Benzene and cyclohexane
(2) Ethyne and Propyne
(3) Benzene and Ethyne
(4) Propyne and Cyclohexane

49. Given below are two statements:
Statement I: 2.005 has four significant figures.
Statement II: Zeros preceding to first non-zero digit are significant.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both statement I and statement II are incorrect
(2) Both statement I and statement II are correct
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

50. Which of the following pairs of compounds follow law of multiple proportions?
(1) H_2O and Na_2O (2) CO_2 and CO
(3) CO_2 and NO_2 (4) BaO and BeO

51. Incorrect statement regarding Dalton's Atomic theory is
(1) Matter consists of indivisible atoms
(2) All the atoms of a given element have identical properties
(3) Compounds are formed when atoms of same elements combined in a fixed ratio
(4) Chemical reactions involve reorganisation of atoms

52. If mass percentage of zinc in a biomolecule is 0.4% then the minimum molecular mass of the biomolecule will be (Atomic mass of zinc = 65.4 u)
(1) 12250 u (2) 22275 u
(3) 16350 u (4) 6250 u

53. In a 0.2 molal aqueous solution of glucose if amount of glucose is 1.8 g, then mass of solvent will be
(1) 45.2 g (2) 55 g
(3) 48.2 g (4) 50 g

54. Suppose the elements A and B combine to form two compounds A_2B and A_2B_3. When 0.25 mol of A_2B weighs 20 g and 0.1 mol of A_2B_3 weighs 10 g, then the atomic weights (in u) of A and B respectively are
(1) 35, 20 (2) 20, 25
(3) 35, 10 (4) 10, 15

55. An element E has following three isotopes ^{100}E : 80%, ^{99}E : 16%, ^{101}E : 4%.
The average atomic mass of element E is approximately
(1) 97.9 (2) 96.8
(3) 98.8 (4) 99.9

56. Given below are two statements:
Statement I: Equal masses of O_2 and N_2 contain equal number of atoms.
Statement II: Molality and mole fraction are temperature independent quantities.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

57. Match List I with List II and choose the correct option.

	List I (Amount of Substance)		List II (Number of oxygen atoms)
a.	18 g $C_6H_{12}O_6$	(i)	4 N_A
b.	44.8 L NO_2(g) at STP	(ii)	0.03 N_A
c.	1.225 g $KClO_3$	(iii)	0.5 N_A
d.	11.2 L CO(g) at STP	(iv)	0.6 N_A

(1) a(iv), b(i), c(ii), d(iii) (2) a(i), b(ii), c(iii), d(iv)
(3) a(iv), b(iii), c(ii), d(i) (4) a(ii), b(iii), c(i), d(iv)

58. Molarity of 29% $\left(\frac{w}{v}\right)$ H_2SO_4 solution whose density is 1.22 g ml^{-1}, is
(1) 1.8 M (2) 3.6 M
(3) 2.4 M (4) 1.2 M

59. The amount of zinc required to produce 1.12 ml of H_2 at STP on treatment with dilute HCl will be
(1) 65 g (2) 0.065 g
(3) 32.5×10^{-4} g (4) 6.5 g

60. An organic compound containing C, H and O gave the following analysis, C = 40%, H = 6.7% and O = 53.3%. Its empirical formula would be
(1) CHO_2 (2) CH_2O
(3) CHO (4) C_2H_2O

61. 200 mL of $\frac{M}{4}$ HCl solution was diluted to make it $\frac{M}{10}$HCl solution. Volume of H_2O needed for dilution is
(1) 200 mL (2) 300 mL
(3) 500 mL (4) 700 mL

62. Molality of an aqueous solution having 0.9 mole fraction of solvent is
(1) 6.17 m (2) 4.25 m
(3) 3.10 m (4) 5.11 m

63. If 56 g N_2 and 64 g CH_4 are mixed with 80 g SO_3 gas, then mole fraction of CH_4 is (consider gases as non reacting)
(1) 0.52 (2) 0.57
(3) 0.42 (4) 0.24

64. Volume of CO_2 gas liberated at STP by burning 24 g of carbon in excess of air is
(1) 11.2 L (2) 22.4 L
(3) 33.6 L (4) 44.8 L

65. Number of urea molecules present in 400 mL of 2.5 M aqueous solution of urea is
(1) 3.01×10^{22} (2) 6.02×10^{23}
(3) 6.02×10^{26} (4) 3.01×10^{24}

66. Mass of 80% pure $CaCO_3$ required to neutralise 100 mL of 0.5 M HCl solution will be
(1) 2.06 g (2) 3.12 g
(3) 4.15 g (4) 4.50 g

67. Choose the incorrect statement about canal rays
(1) They possess positive charge
(2) They travel in a straight line in the absence of electrical or magnetic field
(3) They emerge from anode and move towards the cathode
(4) Properties of canal rays depend upon the nature of the gas present in cathode ray tube

68. Charge/mass ratio is maximum for
(1) Electron (2) Proton
(3) Neutron (4) α-particle

69. The wave number associated with an electromagnetic radiation with frequency 4×10^{-12} Hz will be
(1) 2.49×10^{-16} m^{-1} (2) 2.16×10^{-15} m^{-1}
(3) 1.33×10^{-20} m^{-1} (4) 1.75×10^{-18} m^{-1}

70. A 100 watt bulb emits monochromatic light of wavelength 400 nm. The number of photons emitted per second by the bulb is
(1) 1×10^{15} (2) 2×10^{20}
(3) 2×10^{10} (4) 8×10^{20}

71. According to Rutherford's nuclear atomic model, the ratio for the radius of the atom to that of the nucleus is about:
(1) 10^5 (2) 10^3
(3) 10^{-5} (4) 10^{-3}

72. According to the Rutherford's nuclear model of atom
(a) Electrons and the nucleus are held together by electrostatic forces of attraction
(b) The positive charge and most of the mass of the atom was densely concentrated in extremely small region called nucleus
(c) The nucleus is surrounded by electrons that move around the nucleus with a very high speed in circular paths called orbits
The correct statement(s) is/are
(1) (b) and (c) only (2) (a), (b) and (c)
(3) (a) and (c) only (4) (a) only

73. The longest wavelength in the Lyman series of the atomic hydrogen spectrum is [R_H = Rydberg constant]
(1) $\frac{1}{3R_H}$ (2) $\frac{4}{3R_H}$
(3) $\frac{3R_H}{4}$ (4) $\frac{R_H}{2}$

74. The correct decreasing order of wavelength of given electromagnetic radiations is
(1) Radio waves > Microwave > UV > IR
(2) UV > IR > Microwave > Radio waves
(3) Radio waves > Microwave > IR > UV
(4) IR > UV > Microwave > Radio waves

75. The maximum kinetic energy of photoelectrons emitted when a light of frequency 1×10^{16} Hz is irradiated on a metal surface with threshold frequency (ν_0) = 2×10^{15} Hz is
(1) 3.6×10^{-18} J (2) 1.7×10^{-20} J
(3) 7.5×10^{-19} J (4) 5.3×10^{-18} J

76. Given below are two statements.

Statement I : The radius of 2nd orbit of He^+ ion is 1.058 Å.
Statement II : Splitting of spectral line in electrical field is called Stark effect.
In light of above statements, choose the **correct** answer.
(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are incorrect
(4) Both statement I and statement II are correct

77. Match List I with List II and choose the correct option.

List I (Metals)	List II (Work function/eV)
(a) Li	(i) 2.25
(b) Na	(ii) 2.42
(c) K	(iii) 3.7
(d) Mg	(iv) 2.3

(1) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)
(2) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)
(3) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)
(4) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

78. Match List-I with List-II.

	List I (Electronic transition)		List II (Maximum number of emission lines in H atom sample)
a.	$n_2 = 5$ to $n_1 = 1$	(i)	3
b.	$n_2 = 3$ to $n_1 = 2$	(ii)	10
c.	$n_2 = 4$ to $n_1 = 2$	(iii)	1
d.	$n_2 = 6$ to $n_1 = 1$	(iv)	15

Choose the correct option.
(1) a(i), b(ii), c(iii), d(iv)
(2) a(iv), b(iii), c(ii), d(i)
(3) a(iii), b(i), c(iv), d(ii)
(4) a(ii), b(iii), c(i), d(iv)

79. The pair of series of spectral lines for hydrogen atom which fall in UV and visible region respectively are
(1) Lyman and Pfund
(2) Paschen and Brackett
(3) Balmer and Brackett
(4) Lyman and Balmer

80. According to Bohr's theory angular momentum of an electron in 6th orbit is
(1) $2.5\,\frac{h}{\pi}$
(2) $6\frac{h}{\pi}$
(3) $3\,\frac{h}{\pi}$
(4) $\frac{2.5h}{2\pi}$

81. The ratio of energy of electron in 1st excited state of Li^{2+} ion to the energy of electron in 2nd excited state of He^+ ion is
(1) 81 : 16
(2) 16 : 81
(3) 9 : 4
(4) 4 : 9

CC-001

82. If radii of 3rd orbit of He^+ ion is x Å then the radii (in Å) of same orbit of Li^{2+} ion will be
(1) x
(2) $\frac{x}{2}$
(3) $\frac{3x}{2}$
(4) $\frac{2x}{3}$

83. The energies E_1 and E_2 of two monochromatic radiations are 5 eV and 10 eV respectively. The relation between their wavelengths λ_1 and λ_2 will be
(1) $\lambda_1 = 2\lambda_2$
(2) $\lambda_1 = \lambda_2$
(3) $\lambda_1 = \frac{\lambda_2}{2}$
(4) $4\lambda_1 = \lambda_2$

84. If a particle of mass 0.01 g is moving with a velocity of 200 m/s then the de-Broglie wavelength associated with the moving particle will be ($h = 6.6 \times 10^{-34}$ Js)
(1) 3.3×10^{-34} m
(2) 3.3×10^{-30} m
(3) 3.3×10^{-32} m
(4) 3.3×10^{-31} m

85. Number of electrons in nitrogen atom for which azimuthal quantum number(l) is 1, is
(1) 7
(2) 3
(3) 4
(4) 6

86. Orbital angular momentum of electron present in a 3p orbital is
(1) $\sqrt{12}\hbar$
(2) $\sqrt{6}\hbar$
(3) $\sqrt{2}\hbar$
(4) $4\hbar$

87. The incorrect representation for the given set of quantum numbers is

	n	l	m_l	s
(1)	4	2	–2	+1/2
(2)	3	2	–1	–1/2
(3)	2	1	–2	+1/2
(4)	5	3	0	+1/2

(1) (1)
(2) (2)
(3) (3)
(4) (4)

88. Match the following columns I and II.

	Column I (Element)		Column II (Number of unpaired electron(s))
a.	Copper	(i)	5
b.	Manganese	(ii)	4
c.	Chromium	(iii)	1
d.	Iron	(iv)	6

Choose the correct option.
(1) a(ii), b(iv), c(i), d(iii)
(2) a(ii), b(iii), c(iv), d(i)
(3) a(iii), b(i), c(ii), d(iv)
(4) a(iii), b(i), c(iv), d(ii)

89. Number of degenerate orbitals of hydrogen atom in the 4th shell are
(1) 6
(2) 9
(3) 12
(4) 16

90. Maximum possible electrons in Mn for which n + l = 4 are
(1) 8
(2) 5
(3) 13
(4) 2

BOTANY

91. Which of the following stage lasts for many years in oocytes of some vertebrates?
(1) Diakinesis
(2) Leptotene
(3) Zygotene
(4) Diplotene

92. Mesosomes in a bacterial cell
(1) Are actually infoldings of plasma membrane.
(2) Help in DNA replication only.
(3) Do not contribute in cell wall formation.
(4) Have its own DNA and RNA

93. For the formation of cell plate in plant cells during cytokinesis, the responsible organelle is
(1) Peroxisome
(2) Golgi complex
(3) Smooth Endoplasmic Reticulum
(4) Lysosome

94. Microbodies
(1) Are membraneless
(2) Contain various enzymes
(3) Are present only in animal cell
(4) Are site for protein synthesis in prokaryotes

95. In unicellular organisms such as Amoeba, the functions such as osmoregulation and excretion is associated with
(1) Endoplasmic reticulum
(2) Golgi complex
(3) Contractile vacuole
(4) Gas vacuole

96. The lowest category that is common to both potato and wheat is
(1) Class
(2) Family
(3) Division
(4) Order

97. Match the following columns and select the **correct** option.

Column I	Column II
a. Chloroplast	(i) Cristae
b. Plasmodesmata	(ii) Thylakoid
c. Mitochondria	(iii) Calcium pectate
d. Middle lamella	(iv) Connects cytoplasm of adjacent cells

(1) a(ii), b(iv), c(i), d(iii)
(2) a(ii), b(i), c(iii), d(iv)
(3) a(iii), b(iv), c(ii), d(i)
(4) a(iv), b(i), c(ii), d(iii)

98. Na^+/K^+ pump in the plasma membrane
(1) Performs active transport of ions across the membrane
(2) Transports the ions through transporter proteins only along the concentration gradient
(3) Is responsible to generate ATP molecules from their components
(4) Does not require any energy as it occurs through integral proteins

99. Local names of various plants and animals
(1) Help in recognising organisms worldwide
(2) Are used universally
(3) Are specific and distinct names
(4) Vary from place to place

100. Arthropoda is a/an
(1) Order
(2) Taxon
(3) Class
(4) Kingdom

101. Tonoplast is the
(1) Single membrane around lysosome
(2) Double membrane complex around lysosome
(3) Single membrane around sap-vacuole
(4) Double membrane complex around sap-vacuole

102. Select the **incorrect** statement.
(1) Egg of ostrich is the largest isolated single cell
(2) Ribosomes is membraneless organelle
(3) The shape of the cells may vary with the function they perform
(4) Genetic material is present in well defined nucleus in both prokaryotes and eukaryotes

103. Given below are two statements
Statement I: Main arena of various cellular activities of the cell is cytoplasm.
Statement II: Various chemical reactions occur in the cytoplasm to keep the cell in the living state.
Choose the **correct** option
(1) Only statement I is correct
(2) Only statement II is correct
(3) Both the statements I and II are incorrect
(4) Both the statements I and II are correct

104. Which of the following options shows taxonomic hierarchy in descending sequence?
(1) Class – Order – Family – Kingdom
(2) Order – Division – Class – Species
(3) Kingdom – Division – Class – Order
(4) Kingdom – Class – Division – Family

105. Cell theory does **not** state that
(1) All living organisms are composed of cells
(2) A newly formed cell differentiate and become specified for a specific function
(3) All cells arise from pre-existing cells
(4) Activities of an organism are outcome of activities and interactions of its constituent cells

106. Scientific name of mango is **correctly** written as
(1) *Mangifera Indica*
(2) Mangifera Indica
(3) *Mangifera indica*
(4) *mangifera indica*

107. The size of Pleuropneumonia - like organism (PPLO) is
(1) 0.1 µm
(2) 0.02 µm
(3) 1 – 2 µm
(4) 10 – 20 µm

108. Choose the **incorrect** match
(1) White blood cells – Amoeboid structure
(2) Tracheids – Round and oval
(3) Nerve cells – Branched and long
(4) Red blood cells – Round and biconcave

109. Enzyme recombinase is required at which stage of the prophase-I?
(1) Zygotene
(2) Diplotene
(3) Pachytene
(4) Diakinesis

110. Satellite in chromosome represents
(1) Non-staining primary constriction
(2) Staining primary constriction
(3) Staining secondary constriction
(4) The fragment beyond non-staining secondary constriction

111. Select the **odd** one out w.r.t. inclusion bodies.
(1) Glycogen granules
(2) Polysome
(3) Phosphate granules
(4) Cyanophycean granules

112. **Assertion (A) :** The functions of mitochondria and chloroplasts are coordinated with endoplasmic reticulum.
Reason (R) : Mitochondria and chloroplasts are considered as part of the endomembrane system.
In the light of the above statements, choose the **correct** answer from the options given below.
(1) Both (A) & (R) are true and the (R) is the correct explanation of (A)
(2) Both (A) & (R) are true but the (R) is not the correct explanation of the (A)
(3) (A) is true but (R) is false
(4) Both (A) and (R) are false

113. ICBN stands for
(1) International Code for Bacterial Nomenclature
(2) Indian Code for Botanical Nomenclature
(3) International Code for Biological Nomenclature
(4) International Code for Botanical Nomenclature

114. The period of cell cycle in yeast is
(1) 30 min (2) 60 min
(3) 90 min (4) 24 hours

115. Match the following columns and select the **correct** option

	Column I		Column II
a.	Taxonomy	(i)	These are distinct biological entities and not merely morphological aggregates
b.	Systematics	(ii)	A process of classification, based on characteristics, all living organisms can be classified into different taxa
c.	Classification	(iii)	It takes into account evolutionary relationships between organisms
d.	Taxonomic categories	(iv)	Process by which anything is grouped into convenient categories based on some easily observable characters

(1) a(iii), b(iv), c(i), d(ii) (2) a(ii), b(iii), c(iv), d(i)
(3) a(ii), b(i), c(iv), d(iii) (4) a(iv), b(iii), c(i), d(ii)

116. Gibbon and tiger share the same
(1) Family (2) Class
(3) Order (4) Genus

117. Eukaryotic cilia and flagella
(1) Are hair-like outgrowth of cell membrane
(2) Lack microtubules
(3) Have peripheral triplet
(4) Help in spindle fibre formation

118. Splitting of centromeres occurs in
(1) Mitotic anaphase and meiotic anaphase II
(2) Meiotic anaphase I and telophase II
(3) Mitotic metaphase and meiotic anaphase II
(4) Meiotic prophase II and anaphase I

119. During interkinesis stage,
(1) Replication of DNA occurs.
(2) Replication of centrioles occurs in plant cells.
(3) Chromosomes form chromatin fibres.
(4) RNA and proteins are synthesized.

120. Small bristle like structures that are known to help attach the bacteria to rocks in streams and also to the host tissues, are
(1) Flagella (2) Filaments
(3) Fimbriae (4) Polysomes

121. A type of leucoplast which stores oils and fats is
(1) Chromoplast (2) Aleuroplast
(3) Amyloplast (4) Elaioplast

122. The longest phase of cell cycle is
(1) Metaphase (2) Prophase
(3) Interphase (4) Anaphase

123. All of the following are membrane bound cell organelles, **except**
(1) Centriole (2) Golgi complex
(3) Lysosome (4) Mitochondrion

124. If there are 20 chromosomes in G_1 phase then the number of chromosomes in G_2 phase will be
(1) 10 (2) 20
(3) 30 (4) 40

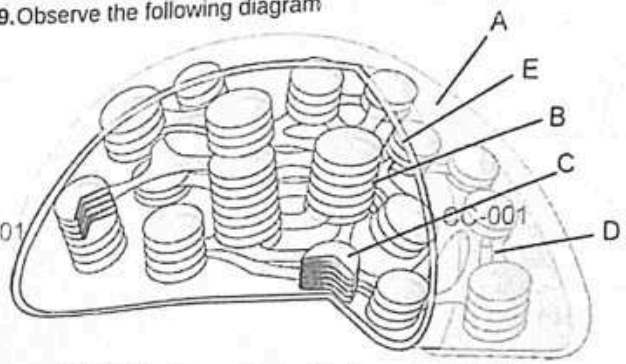
125. All of the given are significance of mitosis, **except**
(1) Growth (2) Repair
(3) Healing and regeneration (4) Introduction of variations

126. Homologous chromosomes align themselves at the equator during
(1) Metaphase-I (2) Prophase-I
(3) Anaphase-II (4) Telophase-II

127. Read the given statements and choose the correct option.
Statement A: Carolus Linnaeus almost single-handedly made the origin of species diversity, the central question of evolutionary biology that it is today.
Statement B: Animals, mammals and dogs represent taxa at different levels.
(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

128. When karyokinesis is **not** followed by cytokinesis, it results in the
(1) Entry of the cell into G_0 phase
(2) Formation of cell plate
(3) Haploid stage of the cell
(4) Formation of syncytium

129. Observe the following diagram



Based on the above figure, select the **correct** option.
(1) Label A is relatively more permeable than Label E
(2) Label B represents flat membranous tubules
(3) Label C represents the space limited by the inner membrane
(4) Label D represents the flattened membranous sacs

130. Higher the category in the taxonomic hierarchy,
(1) More is the number of common characteristics among the organisms
(2) Lesser is the complexity in the classification process
(3) Greater is the difficulty in determining the relationship to other taxa at the same level
(4) Lesser is the number of organisms in that taxa

131. Centriole duplication in animal cells, takes place in:
(1) G_1 phase (2) S phase
(3) G_2 phase (4) G_0 phase

132. Which of the following is **not** a function of mitotic apparatus?
(1) Moving chromosomes to the equator of cell
(2) Replicating DNA
(3) Moving the chromatids towards the poles
(4) Separating sister chromatids

133. Match the following columns and choose the **correct** option.

Column I	Column II
a. Diplotene	(i) Coiling of chromatin begins
b. Zygotene	(ii) Chiasmata become visible
c. Diakinesis	(iii) Nucleolus disappears
d. Leptotene	(iv) Synapsis occurs

(1) a-(iii), b-(i), c-(ii), d-(iv) (2) a-(ii), b-(i), c-(iii), d-(iv)
(3) a-(ii), b-(iv), c-(iii), d-(i) (4) a-(iv), b-(i), c-(ii), d-(iii)

134. Best stage to study morphology of chromosomes is
(1) Metaphase (2) Anaphase
(3) Prophase (4) Telophase

135. How many mitotic divisions are required to produce 256 cells from a single cell?
(1) 255 (2) 256
(3) 7 (4) 8

ZOOLOGY

136. Division of labour by assigning different functions to cells forming different types of tissue was first observed in the phylum
(1) Platyhelminthes (2) Cnidaria
(3) Aschelminthes (4) Porifera

137. Which of the following possesses cells with irregular boundaries that are compactly packed with little intercellular matrix?
(1) Connective tissue (2) Muscular tissue
(3) Epithelial tissue (4) Nervous tissue

138. A group of A cells along with B substances perform a specific function such an organisations is called tissue.
Choose the option that correctly fills the blanks A and B
(1) Dissimilar, Intracellular (2) Similar, Intracellular
(3) Dissimilar, Intercellular (4) Similar, Intercellular

139. Which of the following is not a function of simple epithelium?
(1) Absorption and secretion
(2) Diffusion of respiratory gases
(3) To provide protection against mechanical and chemical stress
(4) Helps in movement of substances in one direction

140. Match items in column I with those in column II and choose the option with all **correct** match.

	Column I		Column II
a.	Chondrocytes	(i)	Greatest control over the body's responsiveness
b.	Ciliated epithelium	(ii)	Present in Buccal cavity
c.	Compound epithelium	(iii)	Present in lacunae of cartilage
d.	Neural tissue	(iv)	Fallopian tubes

(1) a(i), b(ii), c(iii), d(iv) (2) a(iv), b(iii), c(ii), d(i)
(3) a(iii), b(iv), c(ii), d(i) (4) a(ii), b(iv), c(i), d(iii)

141. The type of epithelium present in the inner lining of ducts of salivary gland, is also present in the
(1) Walls of blood vessels
(2) Lining of stomach and intestine
(3) Inner surface of hollow organs like bronchioles
(4) Moist surface of buccal cavity

142. A single thin layer of flattened cells with irregular boundaries make up
(1) Squamous epithelium (2) Cuboidal epithelium
(3) Columnar epithelium (4) Ciliated epithelium

143. A histologist observes an epithelium containing single layer of tall and slender cells, having basal nuclei with mucous secreting cell in between them, on a slide through microscope.
Epithelium being discussed in the above statement is
(1) Simple squamous epithelium
(2) Simple cuboidal epithelium
(3) Stratified squamous epithelium
(4) Columnar glandular epithelium

144. In humans, the epithelium which has limited role in secretion and absorption, does/is not
(1) Cover the dry surface of the skin
(2) Cover the moist surface of buccal cavity
(3) Found in the fallopian tubes
(4) Found in the inner lining of pancreatic ducts

145. Among the following secretions, which one is directly poured into the main circulating fluid of the human body?
(1) Saliva (2) Mucus
(3) Milk (4) Insulin

146. The type of epithelium whose cells bear cilia on their free surface is mainly present in the
(1) Lining of stomach
(2) Inner surface of hollow organs like bronchioles
(3) Tubular parts of nephron in kidneys
(4) Walls of blood vessels

147. The most widely distributed connective tissue in the body of complex animals is
(1) Areolar tissue (2) Tendon
(3) Cartilage (4) Dense irregular tissue

148. Bones are the hematopoietic organs in adult humans. They are actually the type of
(1) Adipose tissue
(2) Areolar tissue
(3) Specialised connective tissue
(4) Fluid connective tissue

149. **Assertion (A):** Cartilage is found more abundantly in vertebrate embryos.
Reason (R): Skeleton of adult vertebrates has no cartilage.
In the light of above statements, choose the correct option.
(1) Both Assertion and Reason are true but Reason is **not** the correct explanation of the Assertion
(2) Both Assertion and Reason are true and Reason is the **correct** explanation of the Assertion
(3) Both Assertion and Reason are false statements
(4) Assertion is true statement but Reason is false

150. Which of the following cells are exclusively present in specialized connective tissue?
(1) Fibroblasts (2) Macrophages
(3) Chondrocytes (4) Mast cell

151. All of the following are present in the main circulating fluid of the body that helps in the transport of various substances, **except**
(1) Erythrocytes (2) Thrombocytes
(3) Macrophages (4) Leucocytes

152. A attach skeletal muscles to bone whereas B attach one bone to another. Here A and B are
(1) A-Ligaments B-Tendons
(2) A-Tendons B-Ligaments
(3) A-Tendons B-Adipose tissue
(4) A-Cartilage B-Ligaments

153. Collagen fibres are compactly packed in the fashion of parallel bundles in
(1) Adipose tissue (2) Areolar tissue
(3) Tendons (4) Dermis of skin

154. Which of the following type of muscles is voluntary and regulated by central nervous system?
(1) Smooth muscle (2) Non-striated muscle
(3) Cardiac muscle (4) Skeletal muscle

155. How many of the below given features belong to smooth muscle fibres?

Fusiform, Multinucleated, Involuntary, no striations, attached to skeletal bones

Select the **correct** option:
(1) Two (2) Three
(3) Four (4) Five

156. The tissue which exerts greatest control over the body's responsiveness to stimuli is
(1) Cardiac muscle tissue (2) Neural tissue
(3) Skeletal tissue (4) Blood

157. All of the following molecules/compounds are found in the acid-soluble pool obtained upon chemical analysis of a living tissue, **except**
(1) Monosaccharides (2) Nucleosides
(3) Amino acids (4) Lipids

158. The category of biomolecule to which inulin and lecithin belong to constitute _____ per cent and _____ per cent of the total cellular mass of a living tissue respectively.
Select the option to fill in the blanks **correctly**.
(1) 7, 10 (2) 3, 2
(3) 3, 5 (4) 5, 7

159. Category of biomolecules that forms 10-15% of the total cellular mass of living tissues are
(1) Carbohydrates (2) Lipids
(3) Nucleic acids (4) Proteins

160. Select the mismatch w.r.t. category of secondary metabolites listed and their example.
(1) Alkaloid – Morphine
(2) Toxin – Carotenoid
(3) Drug – Vinblastine
(4) Essential oil – Lemon grass oil

161. Which of the following happens if zinc is removed from carboxypeptidase?
(1) Its catalytic activity is lost
(2) Its catalytic efficiency increases
(3) It gets denatured
(4) No effect on the activity of enzyme

162. Chitin is present in exoskeleton of arthropods. It is a
(1) Heteropolymer
(2) Polymer of glucose
(3) Polymer of N-acetylglucosamine
(4) Disaccharide

163. Which among the following can hold I_2 molecules in their helical portions and gives blue colour?
(1) Cellulose (2) Starch
(3) Glycogen (4) Glucose

164. In a polysaccharide chain such as glycogen, the right end is called the _____ and the left end is called the _____ respectively.
(1) Reducing end, reducing end
(2) Non-reducing end, non-reducing end
(3) Non-reducing end, reducing end
(4) Reducing end, non-reducing end

165. The biomolecule that **cannot** be hydrolysed further is
(1) Starch (2) Insulin
(3) Lactose (4) Glucose

166. In all of the following compounds, four valency positions of α-carbon are occupied by four different substituent groups, **except**
(1) Glycine (2) Alanine
(3) Serine (4) Valine

167. Variable R-group in amino acid serine is
(1) Methyl (2) Hydrogen
(3) Hydroxy methyl (4) Carboxyl

168. Zwitterionic form is one at which the monomeric form of proteins exists as electrically neutral moiety at a specific pH. This is **true** for
(1) Trihydroxypropane (2) Amino acids
(3) DNA (4) RNA

169. Which of the following elements constitutes negligible percentage weight of human body?
(1) Carbon (2) Nitrogen
(3) Calcium (4) Silicon

170. Which of the following options correctly represents basic amino acid and sulphur containing amino acid, respectively?
(1) Serine, valine (2) Lysine, cysteine
(3) Glycine, lysine (4) Valine, methionine

171. Haemoglobin exhibits optimal activity at which level of structural organisation?
(1) Primary (2) Secondary
(3) Tertiary (4) Quaternary

172. Which of the following bonds is/are present in tertiary structure of a protein?
(1) Covalent, phosphodiester and hydrogen bonds
(2) Peptide bond only
(3) Peptide and glycosidic bonds
(4) Hydrogen, peptide and disulphide bonds

173. The sequence of amino acids i.e., the positional information in a protein-which is the first amino acid, which is second, and so on is called
(1) Tertiary structure (2) Secondary structure
(3) Primary structure (4) Quaternary structure

174. Select the option representing the molecules having peptide bond and glycosidic bond respectively.
(1) Lecithin, Glucagon (2) Insulin, Glycogen
(3) Inulin, Insulin (4) Myoglobin, Collagen

175. Select the correct representation of molecular formula of palmitic acid.
(1) $CH_3-(CH_2)_{12}-COOH$ (2) $CH_3-(CH_2)_{18}-COOH$
(3) $CH_3-(CH_2)_{14}-COOH$ (4) $CH_3-(CH_2)_{16}-COOH$

176. Which one is **true** regarding a coenzyme?
(1) It is always an inorganic compound
(2) It is always a metal ion
(3) It is always a protein
(4) It is often a vitamin

177. An unsaturated fatty acid with 20 carbon atoms is
(1) Guanylic acid (2) Arachidonic acid
(3) Palmitic acid (4) Trichloroacetic acid

178. Number of enzymes have non-protein constituents called co-factors. In these instances, the protein portion of the enzyme is called
(1) Apoenzyme (2) Co-enzyme
(3) Holoenzyme (4) Prosthetic group

179. Competitive inhibition of succinic dehydrogenase occurs by
(1) Succinate (2) Malonate
(3) Melanin (4) Melatonin

180. Read the given statements and select the **correct** option.
Statement A : An active site of a substrate is a crevice or pocket into which the enzyme fits.
Statement B : Enzyme-substrate complex formed is transient and stable.
(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

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